

Data Collection and Preprocessing Phase

Date	4 October 2026
Team ID	N/A
Project Title	Student Mental Health Analysis (Tableau Dashboards & Story)
Maximum Marks	3 Marks

Data Quality Report Template

The Data Quality Report Template will summarize data quality issues from the selected source, including severity levels and resolution plans. It will aid in systematically identifying and rectifying data discrepancies.

Data Source	Data Quality Issue	Severity	Resolution Plan
Dataset	Column name "Attendance_Rate (%)" has a space, brackets and a % sign, which is awkward in code and Tableau calculated fields.	Low	Keep the original header for the raw file. In Tableau, rename the field to "Attendance Rate (%)" and refer to it by that alias. For Python, rename to Attendance_Rate_Pct.
Dataset	Missing values: checked every column and found none (0 nulls).	Low	No imputation or row removal needed. Verified with a null count per column (df.isna().sum()).
Dataset	Duplicates: no duplicate rows and every Student_ID is unique (1,000 of 1,000).	Low	No action needed. Verified with df.duplicated() and Student_ID uniqueness.
Dataset	Outliers: 12 Sleep_Duration_Hours values (2.84 to 10.37 h) and 2 Heart_Rate_Variability values (36.6 and 99.21) fall outside 1.5 x IQR.	Low	Keep them. The values are physiologically plausible, so removing them would only bias the averages. They can be reviewed on a

			box plot.
Dataset	Student_ID and Age are stored as numbers, so Tableau treats them as measures and would sum or average them.	Low	Treat Student_ID as an identifier and set Age as a Dimension (category) so it can be used in Age vs Risk.
Dataset	The journal field is free text but has only 10 distinct sentences repeated across 1,000 rows, so real text analysis is not possible.	Moderate	Treat it as a categorical field (10 labels) and analyse counts and percentages instead of text mining.
Dataset	Personalized_Intervention_Strategy does not follow Mental_Health_Risk: the same strategies appear in every risk level.	Moderate	Do not draw conclusions about which strategy works. Analyse it as a separate category and state this limit in the report.
Dataset	The data looks synthetic: values are spread almost evenly and averages are nearly identical across gender and risk level, so no single factor predicts risk.	High	Keep the data, but present findings as illustrative, not real-world proof. The Story ends with the takeaway that broad, inclusive support fits better than targeting one group.